

Exercise2

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Exercise 2

10 marbles are in a bag. 3 are red, 4 blue and 3 yellow.

Two marbles are drawn without replacement. Calculate the probability that colors are red and blue. (Order does not matter)

(Ans. 0.27)

#Solution here

Case 1: red then blue

$$P_1 = \frac{3}{10} \times \frac{4}{9}$$

Case 2: blue then red

$$P_2 = \frac{4}{10} \times \frac{3}{9}$$

$$P = P_1 + P_2 = \frac{3}{10} \cdot \frac{4}{9} + \frac{4}{10} \cdot \frac{3}{9} = \frac{24}{90} = \frac{4}{15} \approx 0.27$$